Supplemental Material: Strange Attractors in Gradient Descent: Data Structure and Loss Geometry Control Fractal Dimension

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PERTURBATION SENSITIVITY ANALYSIS

The perturbation scale ε used in Lyapunov exponent computation must be chosen carefully. We performed a sensitivity sweep over $\varepsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}, 10^{-7}, 10^{-8}, 10^{-9}\}$ at five learning rates for the baseline MLP (14K parameters, synthetic data, 5 seeds each).

At $\varepsilon = 10^{-8}$ and below, the measured Lyapunov exponent is uniformly $\lambda \approx +0.00004$ per training step regardless of learning rate—a numerical artifact arising from finite-precision arithmetic rather than genuine dynamical sensitivity. Converged measurements require $\varepsilon \geq 10^{-6}$. All results in the main text use $\varepsilon = 10^{-5}$.

This finding has implications for prior work: Lyapunov exponent measurements at $\varepsilon = 10^{-8}$ or smaller should be interpreted with caution, as the reported positive exponents may reflect numerical noise rather than chaotic dynamics.

CORRELATION DIMENSION PIPELINE CALIBRATION

We validated our D_2 pipeline against systems with known correlation dimensions, using the same trajectory length (∼400 points after transient removal) and algorithm parameters as the neural network experiments. Table I summarizes the results.

Two systematic biases are visible. Smooth manifolds are overestimated at higher dimensions (the 3-torus reads 3.74 instead of 3.0). Fractal attractors are underestimated, with bias growing with the true dimension, peaking around $D_2 \approx 5$ where the Mackey-Glass τ=30 system (expected 5.0) reads 3.17. This means our reported D_2 values for the CNN/CIFAR-10 attractor ($D_2 = 3.67$) are conservative lower bounds; the true fractal dimension is likely 15–35% higher.

TABLE I. D_2 pipeline calibration on known dynamical systems. Measured at $n = 400$ points, matching the production trajectory length used in the neural network experiments. Smooth manifolds are slightly overestimated; fractal attractors are underestimated, with bias growing with true dimension. Values are stable across $n \in \{200, 400, 800, 1600, 3200\}$ (see D_2(N) convergence analysis below).

System	Type	Expected D_2	Measured D_2
1-torus	smooth	1.0	1.02
2-torus	smooth	2.0	2.18
3-torus	smooth	3.0	3.73
4-torus	smooth	4.0	5.22
Hénon map	fractal	1.21	1.16
Rössler	fractal	1.99	1.72
Lorenz	fractal	2.05	1.73
Mackey-Glass τ=17	fractal	2.1	1.93
Mackey-Glass τ=23	fractal	3.6	2.44
Mackey-Glass τ=30	fractal	5.0	3.17
Mackey-Glass τ=50	fractal	7.0	6.03

EXTENDED D_2 DATA

Table II provides D_2 values across all learning rate fractions for the five experimental conditions.

Three features are visible. First, the CNN on CIFAR-10 peaks early (30% EoS) and declines toward $D_2 \approx 2$ at 95% EoS, while both MLPs on CIFAR-10 rise monotonically and peak at 90% EoS. Second, both synthetic conditions remain at $D_2 \approx 0.8$–1.0 across all learning rates, with standard deviations below 0.04. Third, cross-seed variance is largest at the transition: the CNN at 20% EoS shows $D_2 = 2.63 \pm 0.62$ and the MLP (269K) at 60% EoS shows $D_2 = 2.70 \pm 0.41$, reflecting sensitive dependence on initial conditions near the onset of multi-dimensional dynamics. With N=10 seeds, the MLP variances have widened from the suspiciously tight values seen at N=3 (e.g., MLP 156K at 90% EoS: ±0.10 versus the earlier ±0.02), confirming that the original error bars underestimated true seed-to-seed variability.

PERSISTENT HOMOLOGY DETAILS

Tables III–V report persistent homology feature counts for all three CIFAR-10 conditions. Persistence is computed via RIPSER on PCA-reduced (10-dimensional) function-space trajectories, subsampled to 400 points.

At 5% of EoS, the trajectory is a simple convergence path with zero H_1 features in all architectures. As the learning rate increases, hundreds of loops (H_1) and voids (H_2) appear at all persistence scales. The gap ratio remains near 1.0 throughout—no dominant topological feature emerges—which is the signature of fractal structure

TABLE II. Correlation dimension D_2 (mean ± standard deviation across seeds) for all conditions at each tested learning rate fraction. Learning rates expressed as fractions of $2/\lambda_{\max}$.

% EoS	CNN+C-10 (269K, 10)	MLP+C-10 (269K, 10)	MLP+C-10 (156K, 10)	CNN+synth (269K, 10)	MLP+synth (14K, 20)
5	0.99 ± 0.005	0.97 ± 0.002	0.97 ± 0.001	0.94 ± 0.004	0.93 ± 0.006
10	1.03 ± 0.02	0.98 ± 0.001	0.98 ± 0.001	0.84 ± 0.01	—
15	1.29 ± 0.12	0.98 ± 0.002	0.98 ± 0.001	0.86 ± 0.02	—
20	2.63 ± 0.62	0.98 ± 0.002	0.98 ± 0.001	0.91 ± 0.02	0.90 ± 0.002
30	$\mathbf{3.67 \pm 0.08}$	0.98 ± 0.001	0.97 ± 0.008	0.95 ± 0.004	0.91 ± 0.001
40	3.46 ± 0.09	1.09 ± 0.03	1.13 ± 0.06	0.95 ± 0.01	—
50	3.10 ± 0.11	1.44 ± 0.10	1.74 ± 0.28	0.95 ± 0.01	—
60	2.72 ± 0.10	2.70 ± 0.41	3.24 ± 0.35	0.96 ± 0.01	—
70	2.42 ± 0.07	3.90 ± 0.17	4.08 ± 0.12	0.97 ± 0.04	—
80	2.15 ± 0.06	4.28 ± 0.07	4.40 ± 0.11	0.98 ± 0.03	—
90	2.00 ± 0.06	$\mathbf{4.37 \pm 0.05}$	$\mathbf{4.61 \pm 0.10}$	0.98 ± 0.03	—
95	1.92 ± 0.04	—	—	—	—

TABLE III. Persistent homology feature counts (mean across 3 seeds) for CNN/CIFAR-10. Gap ratio is the ratio of the longest to second-longest H_1 lifetime.

% EoS	H_1 features	H_2 features	H_1 gap ratio
5	0	0	—
15	107	50	1.24
20	233	173	1.11
30	386	355	1.16
40	421	401	1.07
60	378	305	1.05
90	299	163	1.07

TABLE IV. Persistent homology feature counts (mean across 3 seeds) for MLP/CIFAR-10, 156K parameters (width 50).

% EoS	H_1 features	H_2 features	H_1 gap ratio
5–25	0	0	—
30	1	0	—
40	77	17	1.24
50	181	92	1.25
60	255	163	1.09
70	313	233	1.09
80	360	308	1.01
90	389	337	1.06

rather than a smooth torus. A clean 2-torus would show exactly 2 dominant H_1 features with a gap ratio $\gg 1$, clearly distinguishable from the remaining topological noise. We note that at 400 points, the high feature counts (up to 392 H_1 features) likely include noise-generated features; the key diagnostic is the gap ratio near 1.0 and the qualitative transition from zero to many features, rather than the absolute count. A proper null model—e.g., persistence computed on a random point cloud of the same size and ambient dimension—would help quantify how many features exceed the noise floor; we leave this for future work. The present analysis should be interpreted as qualitative topological confirmation of fractal structure rather than a precise count of independent topological features.

TABLE V. Persistent homology feature counts (mean across 3 seeds) for MLP/CIFAR-10, 269K parameters (width 85).

% EoS	H_1 features	H_2 features	H_1 gap ratio
5–30	0	0	—
40	42	4	1.10
50	137	57	1.14
60	217	132	1.11
70	307	240	1.10
80	358	297	1.34
90	392	315	1.11

The MLP conditions confirm that persistent homology tracks D_2 across architectures. Both MLPs show zero H_1 features below 30% EoS (where $D_2 \approx 1$), onset of topological complexity at 40% EoS (where D_2 first exceeds 1.0), and hundreds of features at 90% EoS (where $D_2 > 4$). The topological transition occurs at the same learning-rate threshold as the geometric transition measured by correlation dimension. At 90% EoS, both MLPs show comparable feature counts to the CNN at its peak (30% EoS: 386 H_1, 355 H_2), consistent with both architectures accessing the same attractor complexity when driven by the same data.

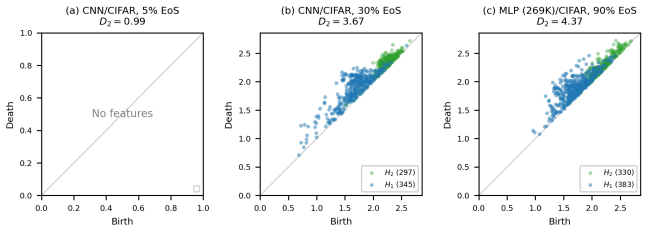

FIG. 1. Representative persistence diagrams. CNN/CIFAR-10 at 5% EoS (trivial: zero features), CNN/CIFAR-10 at 30% EoS (strange attractor: features at all scales), MLP/CIFAR-10 (269K) at 90% EoS (strange attractor: comparable topological complexity to CNN despite different architecture). Points near the diagonal are short-lived; points far from it are persistent topological features.

D_2 CONVERGENCE WITH TRAJECTORY LENGTH

Figure 2 shows the measured D_2 as a function of trajectory length N for three neural network conditions alongside two reference systems (Lorenz attractor, Mackey-Glass τ=30). Longer trajectories were obtained by recording function-space outputs at every training step rather than every 10th step, within the standard 5,000-step protocol; this yields ∼4,000 post-transient points without altering the dynamics.

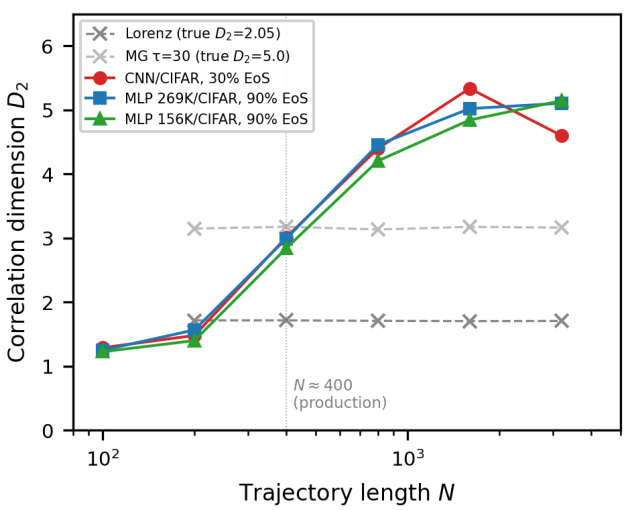

FIG. 2. D_2 versus trajectory length N. Reference systems (gray) are flat: Lorenz at $D_2 \approx 1.72$ and MG τ=30 at $D_2 \approx 3.17$ across all N, confirming pipeline stability. Neural conditions rise from N=100 to N=800 and approach plateau by N=1600–3200. Both MLPs change by $< 7\%$ from N=1600 to N=3200, stabilizing near $D_2 \approx 5.1$. The dashed line marks N≈400, the production trajectory length; reported D_2 values are conservative lower bounds.

The reference systems confirm that the Grassberger-Procaccia pipeline produces stable D_2 estimates across the full range $N \in [200, 3200]$: the Lorenz attractor varies by $< 1\%$ and Mackey-Glass τ=30 by $< 2\%$. For the neural conditions, D_2 rises from N=100 to N=800 as more attractor structure becomes resolvable, then approaches a plateau. Both MLPs stabilize near $D_2 \approx 5.1$ (1.7% and 6.2% change from N=1600 to N=3200). The CNN shows more variation at long N (one seed), but the key result is that D_2 does not grow unboundedly—the attractor has a finite dimension. At the production length N≈400, all three conditions are on the rising portion of the curve, confirming that reported values are conservative lower bounds on the true attractor dimension.

CONVERGENCE OF D_2 AND λ WITH MEASUREMENT PARAMETERS

We verified that the choices $\varepsilon = 10^{-5}$ and $N_{\text{inputs}} = 100$ used in the main text are within converged regimes. For the CNN on CIFAR-10 at 30% of EoS (seed 0), we swept $N_{\text{inputs}} \in \{25, 50, 100, 200, 400\}$ and $\varepsilon \in \{10^{-4}, 5{\times}10^{-5}, 10^{-5}, 5{\times}10^{-6}, 10^{-6}\}$.

The correlation dimension is stable: D_2 varies by only 0.10 across all N_{inputs} values (range 3.74–3.83) and is completely independent of ε, as expected since ε affects only the perturbation protocol, not the trajectory. The production value $D_2 = 3.74$ at $N_{\text{inputs}} = 100$ sits in the center of this range.

The Lyapunov exponent shows more structure: λ varies by a factor of ∼3 across ε values, with a dip at $\varepsilon = 10^{-5}$. All values remain positive ($\lambda \in [0.8, 2.3]{\times}10^{-4}$ per step), confirming chaos across the full ε range tested. The variation with ε is consistent across all N_{inputs} values, indicating a genuine scale-dependent feature of the dynamics rather than a numerical artifact. Crucially, the qualitative conclusion—positive Lyapunov exponent at 30% of EoS—is robust to the choice of ε within $[10^{-6}, 10^{-4}]$.

DEPTH SCALING

We tested MLP depth scaling with tanh networks of width 50 on synthetic data (Table VI).

At depth 4 and 5, chaos persists across the entire tested learning rate range—the "basin wins" regime from the 2-layer MLP disappears. Off-axis dynamics grow to 13% of variance. However, D_2 remains below 1 at all depths on synthetic data, confirming that architectural depth alone is insufficient for multi-dimensional chaos without complex data.

TABLE VI. Chaos window properties versus MLP depth. D_2 remains below 1.0 for all depths on synthetic data, but the chaos window widens with depth as Ruelle-Takens predicts. Peak λ is in units of inverse training steps, matching the main-text convention.

Depth	Params	Window width	Peak λ	Max PC2%
2	14K	0.038	$+1.35 \times 10^{-4}$	5.2
3	17K	0.046	$+2.36 \times 10^{-4}$	8.7
4	19K	0.145	$+3.12 \times 10^{-4}$	12.6
5	22K	0.137	$+2.36 \times 10^{-4}$	12.9

LABEL-NOISE SWEEP

To test whether data complexity acts as a continuous control parameter rather than a binary switch, we sweep label-noise fraction p from 0 (clean CIFAR-10) to 1 (uniformly random labels) at each architecture's peak-D_2 learning rate. A deterministic random number generator (seed 137) selects which samples are relabeled, so the noise pattern is identical across training seeds. The EoS threshold is measured once on clean data ($p = 0$) and held fixed. Seven seeds per noise level.

TABLE VII. D_2 versus label-noise fraction p for two conditions. CNN at 30% EoS: D_2 falls as label structure is degraded. MLP 269K at 90% EoS: D_2 rises modestly as random labels roughen the loss surface.

	CNN/CIFAR (30% EoS)		MLP 269K (90% EoS)	
p	D_2	σ	D_2	σ
0.00	3.64	0.08	4.38	0.07
0.10	3.63	0.12	4.43	0.08
0.25	3.52	0.18	4.54	0.05
0.50	3.09	0.60	4.65	0.08
0.75	2.55	0.71	4.73	0.06
0.90	2.35	0.74	4.69	0.07
1.00	2.34	0.75	4.70	0.09

The CNN result is clear: structured labels create the oscillatory modes that produce multi-dimensional chaos; randomizing them weakens these modes and reduces D_2 by a factor of 1.6. Notably, D_2 at $p = 1$ (2.34 ± 0.75) remains well above the CNN/synthetic baseline ($D_2 \approx 1$), indicating that the geometric structure of the CIFAR-10 inputs contributes to multi-dimensional dynamics even when labels carry no information. The CNN variance grows with noise (σ rises from 0.08 to 0.75), consistent with individual seeds responding heterogeneously as label structure is degraded.

The MLP result is subtler. At 90% of EoS, the dynamics are already near the stability boundary. Random labels create a rougher loss surface—more competing local minima, steeper curvature ridges—that sustains high-dimensional chaos even in the absence of label structure. The rise is modest ($\Delta D_2 \approx 0.3$), concentrated between $p = 0$ and $p = 0.75$, with D_2 plateauing thereafter.

This regime dependence is interpretatively important: "data complexity controls D_2" is accurate at moderate learning rates where the data creates the modes. Near the EoS, the relationship is more nuanced—the landscape roughness produced by incoherent labels provides an alternative source of high-dimensional dynamics.

λ–D_2 DISSOCIATION ACROSS ARCHITECTURES

The main text reports that peak Lyapunov exponent λ and peak correlation dimension D_2 occur at different learning-rate fractions—a dissociation characteristic of the KAM transition. Figure 3 shows the full $\lambda(\eta)$ and $D_2(\eta)$ curves for all three CIFAR-10 conditions, confirming that this dissociation is universal.

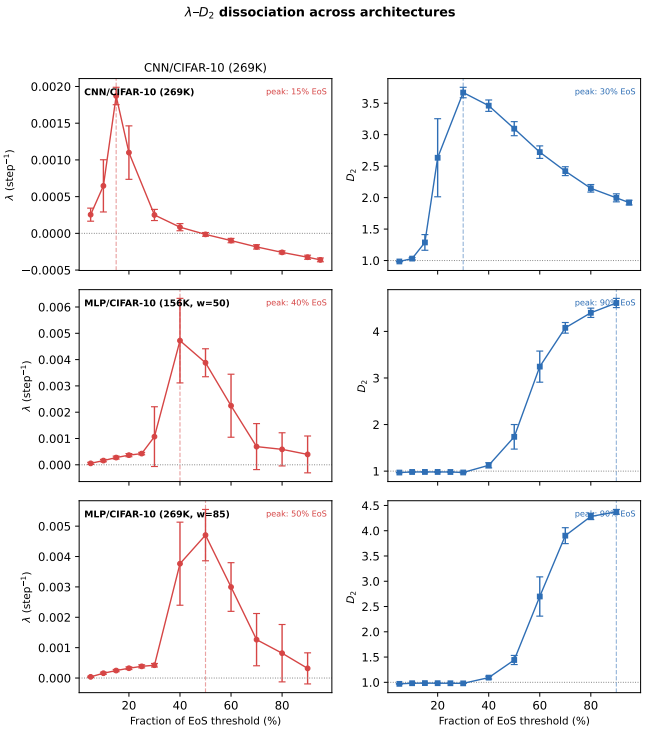

FIG. 3. λ (left) and D_2 (right) versus fraction of the EoS threshold for all three CIFAR-10 conditions. Dashed vertical lines mark each quantity's peak. In every case, peak λ precedes peak D_2: CNN at 15% vs 30%, MLP 156K at 40% vs 90%, MLP 269K at 50% vs 90%. Error bars: standard deviation across 10 seeds.

The dissociation gap varies with architecture: 15 percentage points for the CNN, 50 for the MLP 156K, and 40 for the MLP 269K. In each case, geometric complexity peaks not where perturbation sensitivity is greatest but in the transition zone where chaotic and basin-convergent

dynamics compete. This is consistent with KAM theory, where the richest fractal structure emerges at the boundary between ordered and chaotic phase-space regions.

BATCH-SIZE SWEEP: ROBUSTNESS UNDER SGD

The main text reports all measurements under full-batch gradient descent. To test whether the strange attractor survives stochastic gradient noise, we sweep batch size $B \in \{2000, 1000, 500, 200, 100\}$ for the CNN/CIFAR-10 condition at 30% of EoS, using a paired-noise protocol: at each training step t, a deterministic index generator (seeded by seed $\times\ 10^5 + t$) selects the mini-batch, and both the original and perturbed model copies train on the identical subset. This ensures the Lyapunov exponent measurement isolates deterministic sensitivity from stochastic forcing—the standard paired-noise technique from stochastic dynamical systems. The EoS threshold is measured once at full-batch ($B = 2000$) and the learning rate is held fixed across all batch sizes; only the gradient noise level changes.

Figure 4 and Tables VIII–IX summarize the results. Three seeds per batch size.

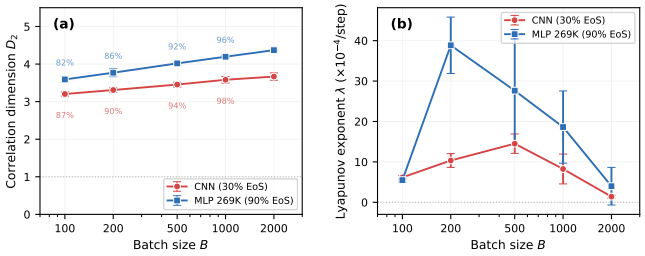

FIG. 4. Batch-size sweep for CNN/CIFAR-10 at 30% EoS (red circles) and MLP 269K at 90% EoS (blue squares). (a) D_2 versus batch size B. Percentages: retention relative to full-batch. (b) λ versus B. SGD amplifies λ by up to 10×. Three seeds per condition; error bars show standard deviation. With only 3 seeds, the standard deviation itself carries ∼50% relative uncertainty; these sweeps are best interpreted as demonstrating qualitative robustness rather than precise quantitative trends.

TABLE VIII. Batch-size sweep for CNN/CIFAR-10 at 30% EoS. D_2 and λ are mean ± standard deviation across 3 seeds. Retention is $D_2(B)/D_2(B{=}2000)$.

B	D_2	λ ($\times 10^{-4}$/step)	PC1%	Retention
2000	3.67 ± 0.10	1.4 ± 0.4	52.1	100%
1000	3.58 ± 0.09	8.2 ± 3.7	52.1	98%
500	3.45 ± 0.06	14.5 ± 2.4	52.0	94%
200	3.31 ± 0.06	10.4 ± 1.7	52.4	90%
100	3.20 ± 0.05	6.2 ± 0.5	53.3	87%

TABLE IX. Batch-size sweep for MLP/CIFAR-10 (269K) at 90% EoS. Same protocol as Table VIII.

B	D_2	λ ($\times 10^{-4}$/step)	PC1%	Retention
2000	4.37 ± 0.01	4.0 ± 4.7	36.0	100%
1000	4.20 ± 0.03	18.6 ± 8.9	36.7	96%
500	4.02 ± 0.02	27.6 ± 13.4	36.0	92%
200	3.77 ± 0.10	38.8 ± 7.0	37.1	86%
100	3.59 ± 0.08	5.5 ± 0.5	37.7	82%

Both architectures show the same qualitative behavior, confirming that the robustness to SGD noise is not architecture-specific. Three features are notable. First, D_2 degrades gracefully in both cases: at $B = 100$, where the gradient is computed on only 5% of the training set per step, the CNN retains 87% and the MLP 82% of their full-batch fractal dimension. Both remain well above $D_2 = 3$. Second, the Lyapunov exponent increases by an order of magnitude under SGD before declining at the smallest batch sizes. For the CNN, λ peaks at $B = 500$ (14.5×10^{-4}, a 10-fold increase over full-batch); for the MLP, λ peaks at $B = 200$ (38.8×10^{-4}, also ∼10× full-batch). This amplification is consistent with stochastic forcing adding energy to the chaotic modes without creating new ones—the number of dynamical dimensions (D_2) changes modestly while the sensitivity along existing dimensions (λ) increases dramatically. Third, PC1 variance is essentially unchanged across all batch sizes for both architectures (∼52% for the CNN, ∼36% for the MLP), indicating that the trajectory's shape in function space is stable under SGD noise.

The non-monotonic behavior of λ with B—rising as B decreases, then falling at the smallest batch sizes—likely reflects two competing effects: decreasing B increases gradient variance (amplifying chaos), but also reduces the effective signal-to-noise ratio in each update (weakening the deterministic dynamics that sustain the attractor). The crossover batch size differs between architectures ($B \approx 500$ for the CNN, $B \approx 200$ for the MLP), consistent with the MLP at 90% EoS having stronger underlying deterministic dynamics that tolerate more noise before the signal degrades.

STATIONARITY OF THE MEASUREMENT WINDOW

The D_2 measurements discard the first 20% of each trajectory as transient. However, as Fig. 1(d) of the main text shows, the top Hessian eigenvalue at 95% of EoS is still rising at step 5,000, raising the question of whether the dynamics have reached a stationary regime at the highest learning rates.

At the learning rates where the main quantitative claims are made (CNN at 30% EoS, MLPs at 90%

EoS), the sharpness dynamics show bounded oscillation rather than secular drift, consistent with a quasi-stationary regime within the measurement window. At 95% EoS, where sharpness is clearly non-stationary, the measured D_2 should be interpreted as characterizing the quasi-stationary dynamics of the approach to equilibrium rather than an asymptotic attractor. We note that this caveat applies primarily to the highest EoS fractions (90–95%), which are not the conditions on which our main claims rest; the CNN peak at 30% EoS and the MLP convergence analysis at 90% EoS both show stable sharpness within the measurement window.

A windowed analysis—computing D_2 separately over the first and second halves of the post-transient trajectory—would provide a direct test of stationarity; we leave this for future work.

EMBEDDING DIMENSION

The Grassberger-Procaccia algorithm is applied in a function-space embedding. The trajectory points are the network's output vectors on a fixed set of held-out inputs, giving a natural embedding in $\mathbb{R}^{N_{\text{inputs}} \times N_{\text{classes}}}$ (here $100 \times 10 = 1000$ dimensions). This high ambient dimension ensures that the embedding dimension far exceeds the measured $D_2 \approx 3$–5, satisfying the Whitney/Takens embedding criterion ($m > 2D_2$) without the need for time-delay embedding. The correlation integral $C(r)$ is computed from pairwise distances in this space, and the scaling exponent is extracted from the linear region of $\ln C(r)$ versus $\ln r$. Because the embedding dimension is fixed and much larger than D_2, we do not perform a $D_2(m)$ convergence sweep—this is unnecessary when the embedding is already sufficient. The PCA reduction to 10 dimensions used for persistent homology is applied only to the TDA analysis, not to the D_2 computation.